Robert (Rob) J. Spenceley

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RESEARCH INTERESTS

My research interests revolve around how humans interact with AI and embodied automation, with a focus on supporting the implementation of AI in high-risk and dynamic teaming environments (healthcare, search and rescue, etc.). This involves refining how information is presented, the form factor of AI systems, and the design of automation that enhances collaboration among human team members.

Keywords: Human-AI Teaming, Human-Robot Interaction, Human Factors, Neuroergonomics.

EDUCATION

University of Wisconsin – Madison (UW-Madison) Jan. 2025 – *Present*
Doctor of Philosophy – Industrial and Systems Engineering (Human Factors)
Advisor: Dr. Ranjana K. Mehta
Expected Graduation Date – Fall 2028

Colorado State University – Fort Collins (CSU) Aug. 2021 – Dec. 2024
Bachelor of Science – Psychology (Concentration: Mind, Brain, & Behavior)
Minor: Business Administration

RESEARCH EXPERIENCE

Graduate Research Assistant – Neuroergonomics Lab Jan. 2025 – *Present*
Department of Engineering – Industrial and Systems Engineering
University of Wisconsin-Madison
Principal Investigator: Dr. Ranjana Mehta

Projects: Physical Human-Robot Interaction (pHRI), Trust in Multi-Human Robot Teams

- Conducted experiments using a breadth of bioinstrumentation (EEG, fNIRS, eye-tracking, AMTI Force Plate EMG, & ECG) and a UR10e robotic arm.
- Using methods such as Lab Streaming Layer, Python, & MATLAB to streamline data collection.
- Analysis of EEG data using power spectrum density, corticomuscuular coupling, and hyperscanning analyses, collected from a virtual reality disaster robotics experiment and pHRI.

National Science Foundation (NSF) Research Traineeship (NRT+) Trainee Jan. 2025 – *Present*
Department of Computer Science - Integrating Robotics into the Future of Work (INTEGRATE)
University of Wisconsin-Madison
Principal Investigator: Dr. Bilge Mutlu

Project: Comfort Survey *(CHIRP)*

- Development of a survey to quantify human comfort when working with robots in occupational settings.
- Leading a team of RAs to develop virtual reality experiments.
- Data collection & analysis of survey data using methods like exploratory factor analysis, confirmatory factor analysis, and the Delphi Method.

Lab Manager – Davalos Lab Jul. 2023 – Dec. 2024
Department of Psychology – Cognitive Neuroscience
Colorado State University

Project: Neural Correlates of Prospective Memory

- Training new undergraduate research assistants/graduate students in EEG, overseeing the creation of data collection schedules, ensuring the proper maintenance of lab equipment, & assisting with other undergraduate projects using the lab's data.
- I have conducted over 300 hours of EEG data collection, building on my prior experience as a research assistant.

Lab Manager – Wickens Lab Aug. 2024 – Dec. 2024
Department of Psychology – Cognitive Psychology
Colorado State University

Projects: FIPRED, Augmented Reality Cueing in Visual Search, Germane Load.

- Overseeing data collection schedules and maintaining clear communication with participants throughout the study process.
- Data collection with graduate student studies and a project I developed, FIPRED.
- I have collected over 150 hours of data using the HoloLens 2, Meta Quest 3, and wide-angle displays.

PUBLICATIONS

Spenceley, R. J., Yadav, A., & Mehta, R. K. (Accepted). An Integrated Multimodal Framework for Measuring Physical Team Processes During State Emergency Operations Center Activation. *Sage Journals In Proceedings of the Human Factors and Ergonomics Society Annual Meeting. Sage CA: Los Angeles, CA: SAGE Publications*

Spenceley, R. J., & Mehta, R. K. (2026). When robots reshape teams: Neurodynamic insights into taskwork and teamwork in search and rescue. *Frontiers in Neuroergonomics*, 7. https://doi.org/10.3389/fnrgo.2026.1771753

Spenceley, R., Wickens, C., & De Toledo Piza., B. (*2025*) FIPRED: A study of predictive automated displays for wildfire fighting management. *Sage Journals In Proceedings of the Human Factors and Ergonomics Society Annual Meeting (p.1-7). Sage CA: Los Angeles, CA: SAGE Publications. https://doi.org/10.1177/10711813251364799*

Wickens, C., & Spenceley R. (In Press). The trajectory prediction challenge (TPC): An application of cognitive engineering. In Khalid, H. M. (Eds.), *Helander's compendium of human factors and ergonomics.*

PRESENTATIONS

Spenceley, R. J., Yadav, A., & Mehta, R. K. (2026). *An Integrated Multimodal Framework for Measuring Physical Team Processes During State Emergency Operations Center Activation*. ASPIRE, Reno, Nevada. **- Lecture**

Spenceley, R. J., & Mehta, R. K. (2026) *When robots reshape teams: Neurodynamic insights into taskwork and teamwork in search and rescue.* Neuroergonomics Conference, Boston, Massachusetts. **– Lecture**

Spenceley, R. J., & Mehta, R. K. (2026) *Are You Ready for This? An Evaluation of Motor-Readiness Cortical Potentials While Co-Lifting with an Unreliable Robot Partner.* Neuroergonomics Conference, Boston, Massachusetts; HFES ASPIRE, Reno, Nevada. **– Poster**

Spenceley, R., Wickens, C., & De Toledo Piza, B. (2025). *FIPRED: A study of predictive automated displays for wildfire fighting management.* Human Factors and Ergonomics Society – ASPIRE, Chicago, Illinois. **– Poster**

Spenceley, R., Yadav, A., & Mehta, R. (2025). *EEG theta power varies across roles in mixed human-robot teams.* Human Factors and Ergonomics Society – ASPIRE, Chicago, Illinois. **– Poster**

Spenceley, R., Rice, M., & Davalos, D. (2024). *Components of executive functioning involved in prospective memory*. Rocky Mountain Psychology Association Conference, Denver, Colorado. **– Poster**

EXTRA CIRICULARS

Human Factors & Ergonomics Society – UW-Madison Chapter
Vice President of Events Fall 2025 – Spring 2026
Chapter President Spring 2026 – Present

REVIEWER

IISE Transactions on Occupational Ergonomics and Human Factors 2026

TEACHING EXPERIENCE

Instructor Fall 2024
PSY 292C – Controversial Issues in Psychology

Responsibilities: In my role as an instructor, I created a welcoming environment where students felt comfortable sharing their opinions on challenging topics. My focus during this course was for students to develop a framework that would allow them to dissect difficult psychological topics, debunk common myths, and become masters of APA format. I was also responsible for facilitating class discussions, creating and grading assignments, and adjusting course materials the interests of my 10 students.

Teaching Assistant Spring & Summer 2022, Summer 2024
PSY 252 – Mind, Brain, & Behavior

Responsibilities: In my role as a teaching assistant, I conducted comprehensive exam reviews to clarify concepts important to the course, and that would be important for later psychology courses, integrating knowledge from upper-division psychology courses and lab experience. Additionally, I was tasked with efficiently grading assignments and tests while providing constructive feedback to support student growth and regularly attending class to build strong relationships with courses of roughly 100 students.

Teaching Assistant Fall 2022
PSY 250 – Research Design & Analysis I

Responsibilities: In my role as a teaching assistant, I reviewed and graded approximately 100 student papers with a specific focus on APA formatting, evaluated extra-credit assignments to determine where additional instruction was needed, demonstrated to students how to locate high-quality academic sources, and collaborated with graduate teaching assistants to coordinate teaching strategies.

HONORS & AWARDS

College of Natural Science Dean's List - *CSU* 2022, 2023, & 2024
Chancellor's Opportunity Award - *UW-Madison* 2024

RELEVANT SKILLS

Research Techniques: EEG/fNIRS cap placement/fitting (10-20), EMG, ECG, human motion tracking (XSENSE), eye-tracking (Tobii), HoloLens 2, & Meta Quest 3, computer vision.
Software: R, MATLAB (EEGLAB/ERPLAB), BrainVision, Homer3, Aurora, Noraxon, Tobii, Lab Streaming Layer (LSL), Curry7, PsychoPy, Python, Qualtrics, Shimmer, Prolific.
Certifications: Behavioral/social CITI Training, Good Clinical Practice CITI Training & Delta Program Mentorship Certificate.

MEMBERSHIPS

Human Factors and Ergonomics Society (HFES) – Student Affiliate 2025 – *Present*
Human AI Teaming (HART) Technical Group (HFES) 2026 – *Present*

APPEARS IN

UW College of Engineering Editorial – How do robots impact search-and-rescue teamwork? Engineers look at human brain activity to find out – **4/13/26**
RIT AWARE AI Newsletter – **December 2025**